

Homework 3

GENERAL ASSIGNMENT INSTRUCTIONS regarding the use of AI tools

AI tools like ChatGPT can be a powerful resource for generating ideas and refining your work. However, it is essential to know how to use these tools effectively and critically. Make sure to verify the relevance and accuracy of the answers you receive and be mindful of how AI-generated content aligns with the requirements of the assignment. Your own judgment and understanding are key to producing high-quality work.

Along with your responses, please append a list of any requests made to AI tools, such as ChatGPT, if used during the process of creating your answer.

Problem 1 (Galapagos finches, problem 18 in Chapter 2 of the 3-rd edition of the *Sleuth*):

Note: I provide here a short description of the problem, as access to the textbook is not a requirement for this course.

One of the classic data sets in evolutionary ecology is the long-term data collected by Peter and Rosemary Grant on finches in the Galapagos. The data are provided in ex0218.csv and describe the size of the beak of finches on Daphne Major. The data set includes the beak size of all 751 finches that were measured in 1976 and all 89 finches that were measured in 1978. They measured all these species of finch on the island each year. 1977 was a severe drought year; the only food source was a large, tough seed that was ignored under normal conditions. However, to survive, a bird must have eaten those large, tough seeds.

You are tasked with conducting a statistical analysis of these data to address the potential beak increase. Please answer the following questions using these data and R.

- Draw side-by-side box plots of beak size for the 1976 and 1978 birds. Make sure you include labels or a legend to identify each of the groups with ease.
- Use a two-sample (pooled) T-test to test the hypothesis of no change in beak size. Please report a two-sided p -value and interpret the results within the context of the problem. Show all your work and make sure you specify the distribution and degrees of freedom used.
- Estimate the increase in mean beak size between 1976 and 1978. Also report the standard error of this estimate and a 95% confidence interval, calculated using standard methods discussed in this class.
- Typical life span for these finches is 5-20 years. Does this raise any concerns about the assumption of independence between the two groups?

Problem 2 (Baseball): In Major League Baseball, the two leagues follow different rules about whether the pitcher (the player who throws the ball) also must take a turn batting.

- In the **American League (AL)**, the **designated hitter rule** is used: pitchers do not bat, and a different player (the designated hitter) bats in their place.
- In the **National League (NL)**, pitchers do bat for themselves.

Because of this rule difference, one might wonder *whether there is a difference in the average combined runs scored per game in the two leagues*.

To investigate, researchers looked at only games where both teams were from the same league (so no cross-league games). They collected two samples from the 2006 season (only intra-league games were considered in the samples):

- **American League:** 12 games, average runs per game = 9.8, with a standard deviation of 0.8.
- **National League:** 14 games, average runs per game = 9.4, with a standard deviation of 0.64.

Using this information,

- a) calculate the pooled estimate of the standard deviation for the two samples and
- b) construct a 95% confidence interval for the difference in mean runs scored between the two leagues.

There is no data provided for this problem, all calculations will be completed by hand. Please show all your work and include a brief interpretation of the interval in (b) in the context of this problem.

Problem 3: (Dietary supplement for weight gain)

Suppose researchers want to test whether a new dietary supplement helps hogs gain more weight. To control for natural differences in growth, they use **6 pairs of hogs that are full siblings** (from the same litter). Within each pair, one hog is **randomly assigned** to receive the dietary supplement, while the other hog receives a **placebo supplement** (a supplement that looks the same but does not contain the active ingredient believed to affect weight gain).

The goal is to see whether the *hogs given the dietary supplement gain more weight than their sibling counterparts*.

The data are provided in **hogs.csv**, where each row corresponds to a sibling pair. Researchers are interested in determining whether there is **statistically significant evidence** that the supplement increases weight gain.

You will answer this by working through the following questions, using **R for numerical calculations**.

- a) Should these data be analyzed using a paired T -test or a two sample T -test? Briefly explain your choice.
- b) Estimate the average change in weight (as Treatment-Placebo). Make sure to include units.
- c) Calculate the standard error of the average change in weight. Make sure to include units.
- d) Test the null hypothesis of no change between Treatment and Placebo weights. Report the p -value and an appropriate one-sentence conclusion from the test.
- e) Compute a 90% confidence interval for the mean difference in weight gain and give an interpretation of this interval within the context of this study.

General homework notes:

1. Be sure to check which way around R is computing the difference.
2. For questions requesting numeric answers, copy the appropriate numbers from computer output and report those in your homework submission.
3. Do NOT hand in copies of computer output. You should keep a copy of your code so you can compare with our code, in case you get different results.
4. If you use Rmarkdown (or similar), please do not include the Rmarkdown output (i.e., code + unfiltered results). See advice point 2).